Running the Plots

Step-by-step guide to regenerating every paper figure

The paper figures (Figures 1–7) are written as PDFs into the **figures/** folder; the Figure 8 onset/chronic report is written to **results/onset_chronic_analysis_report.pdf**. The model results behind them are already committed to the repository, so you do **not** need to retrain any models — you only run the figure scripts.

IMPORTANT
Do **not** run `04_temporal_shuffle_test.py` to make the figures. That is the 1,000-permutation null test (~24 hours) and its output is already included in `results/`.

1 Install Git LFS first

Three large files (the GADM map layers and the monthly GDELT parquet) are stored via Git Large File Storage. Install Git LFS **before** cloning, or Figure 1 (the choropleth maps) will break because the map files arrive as tiny pointer stubs.

- **Windows:** download the installer from https://git-lfs.com and run it.
- **macOS:** brew install git-lfs
- **Linux:** sudo apt install git-lfs (or sudo yum install git-lfs)

Then enable it once per machine:

```
git lfs install
```

2 Clone the repository

Git LFS downloads the large files automatically during clone.

```
git clone https://github.com/News-Signal-for-Food-Insecurity/news-signal-africa.git
cd news-signal-africa
```

TIP
Already cloned without LFS? You don't need to re-clone — just fetch the real files in place with `git lfs pull`.

3 Install Python dependencies

Python 3.9 or later is required.

```
pip install -r requirements.txt
```

> **NOTE**
> The geospatial packages (geopandas, shapely) are needed for the Figure 1 maps only and can be awkward to install on Windows. If they fail, every other figure still generates — Figure 1 is simply skipped.

4 Generate the figures

Run the figure scripts from the repository root:

```
python 06_paper_figures.py         # Figures 1-6
python generate_fig7_new.py        # Figure 7a / 7b panels
python 08_onset_chronic_report.py  # Figure 8 (onset vs. chronic)
```

Figures 1–7 land in the **figures/** folder. Script 08 writes its main report to **results/onset_chronic_analysis_report.pdf** (and its component panels to figures/).

5 Verify the output

- Open the **figures/** folder (and **results/** for the Figure 8 report) and confirm the PDFs were updated.
- If Figure 1 is blank or missing, geopandas/shapely is not installed, or Git LFS did not fetch the .gpkg map files (re-run `git lfs pull`).
- Every other figure should render without the geospatial packages.

Troubleshooting

- **ModuleNotFoundError** — re-run `pip install -r requirements.txt` in the active environment.
- **Figure 1 empty / map files tiny** — Git LFS issue; run `git lfs install` then `git lfs pull`.
- **FileNotFoundError on results/** — run from the repository root, not from inside a subfolder.